PES1PG25CA044
Chovutiya Avikumar

1. Trace the following sorting algorithms for this
I, A, M, C, H, O, L, T, E.

a) Bubble sort

pass no	i	i<=n-2	min	j	j<=n-1	A[j] < A[min]	Action	Array
1	0							

pass no	i	i<=n-2	j	j<=n-2-i	A[j+1]<A[j]	Action no	Array
1	0	0<=7(T)	7	7<=7(T)		5	A I C H M L O E T
2	1	1<=7(T)	6			4	A C H I L M E O T
3	2	2<=7(T)	5			1	A C H I L E M O T
4	3	3<=7(T)	4			1	A C H I E L M O T
5	4	4<=7(T)	3			1	A C H E I L M O T
6	5	5<=7(T)	2			1	A C E H I L M O T
7	6	6<=7(T)	1			0	A C E H I L M O T
8	7	7<=7(T)	0			0	A C E H I L M O T

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Pass	comparisons	swaps
1	8	5
2	7	4
3	6	1
4	5	1
5	4	1
6	3	1
7	2	0
8	1	0

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Selection sort

pass no.	i	no. j	Action	array
1	0	8	swap(A, I)	A I M C H O L T E
2	1	7	swap(I, C)	A C M I H O L T E
3	2	6	swap(M, E)	A C E I H O L T M
4	3	5	swap(I, H)	A C E H I O L T M
5	4	4	~~swap(~~ no changes	A C E H I O L T M
6	5	3	swap(O, L)	A C E H I L O T M
7	6	2	swap(O, M)	A C E H I L M T O
8	7	1	swap(T, O)	A C E H I L M O T

number of calls	number of comparisons
1	8
2	7
3	6
4	5
5	4
6	3
7	2
8	1

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c) Quick sort

pass no.	condition	swap	Array
1	A < I	swap(A, I)	A I M C H O L T E
2	M > I	no swap	A I M C H O L T E
3	C < I	swap C M	A I M C H O L T E
4	H < I	swap H, M	A I C H M O L T E
5	O > I	no swap	
6	L > I	no swap	
7	T > I	no swap	
8	E < I	swap E M	A I C H E O L T M
9	pivot swap	swap I E	A E C H I O L T M
10	C < E	swap C, E	A C E H I O L T M
11	H > E	no swap	
12	L < O	swap L, O	A C E H I L O T M
13	T > O	no swap	

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14	m ~~T~~O	swap m, T	ACEHILOMT
15	Pivot swap	swap O, m	ACEHILMOT

Pass	comparisons	no. of swaps
1	8	4
2	7	1
3	6	1
4	5	1
5	4	1
6	3	6
7	2	0
8	1	0

1 13 3 8 15 12 20 5

I A M C H O L T E

1	1	1<=8	I	0	0>=0(T)	I > A (T)	I I M C H O L T E
				-1	-1 >=0(F)		A I M C H O L T E
2	2	2<=8	M	1	1>=0(T)	I > M(F)	A I M C H O L T E
3	3	3<=8	C	2	2>=0(T)	M>C(T)	A I M M H O L T E
				1	1>=0(T)	I > C (T)	A I I M H O L T E
				0	0>=0(T)	A > C (F)	A C I M H O L T E
4	4	4<=8	H	3	3>=0(T)	M> H (T)	A C I M M O L T E
				2	2>=0(T)	I > H (F)	A C I I M O L T E
				1	1>=0(T)	C > H (F)	A C H I M O L T E
5	5	5<=8	O	4	4>=0(T)	M>O (F)	A C H I M O L T E
6	6	6<=8	L	5	5>=0(T)	O>L(T)	A C H I M O O T E
				4	4>=0(T)	M>L(T)	A C H I M M O T E
				3	3>=0(T)	H >L (F)	A C H I L M O T E
7	7	7<=8	T	6	6>=0(T)	O >T (F)	A C H I L M O T E
8	8	8<=8	E	7	7>=0(T)	T >E (T)	A C H I L M O T T
				6	6>=0(T)	O >E (T)	A C H I L M O O T
				5	5>=0(T)	M>E(T)	A C H I L M M O T
				4	4>=0(T)	L> E(T)	A C H I L L M O T
				3	3>=0(T)	I > E (T)	A C H I I L M O T
				2	2>=0(T)	H >E (T)	A C H H I L M O T
				1	1>=0(T)	C>E (F)	A C E H I L M O T

Pass	comparisons
1	2
2	1
3	3
4	3
5	1
6	3
7	1
8	7